

MOHSINA

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SUMMARY

AI focused Computer Science postgraduate with strong hands on experience in machine learning, deep learning, and LLM based systems. Skilled in building end-to-end AI pipelines, including data preprocessing, model training, and explainability. Experienced in Python-based development for scalable AI solutions, with a focus on real-world applications such as medical imaging and on-device AI systems.

EDUCATION

MSc Computer Science with Artificial Intelligence – CUSAT (2024 -2026)

BSc Computer Science – Calicut University (2021 – 2024)

TECHNICAL SKILLS

Programming: Python, SQL, C

Data Analysis: NumPy, Pandas, Matplotlib, Data Cleaning, Feature Engineering

Machine Learning: Scikit-learn, Model Training, Evaluation, Supervised & Unsupervised Learning

Deep Learning: TensorFlow, PyTorch, CNNs, Transfer Learning, Model Optimization

Computer Vision: Image Classification, Segmentation, Data Annotation Basics, Bounding Boxes

AI Systems & LLM Engineering: Retrieval-Augmented Generation (RAG), Vector Databases, LLM Integration, Embedding-based Retrieval

Databases: MySQL, MongoDB, Firebase, NOSQL

Tools: Git, GitHub, Jupyter Notebook, VS Code, Google Colab, Kaggle

EXPERIENCE

AI Full-Stack Engineering Intern – Mellon AI Labs

- Developed AI-based applications using Python, combining machine learning models and LLMs for practical use cases
- Built RAG pipelines in Python to retrieve relevant information and generate more accurate, context aware responses
- Worked on semantic search using embeddings and vector databases, improving how information is retrieved from large datasets
- Handled data preprocessing, text chunking, and embedding generation using Python for better retrieval performance
- Optimized models for faster on-device inference, reducing response time in real-time applications
- Collaborated with the team using Git and agile workflows, contributing to clean and maintainable code

AI/ML Intern – Elevate Labs

- Developed machine learning models using Python, applying preprocessing and feature engineering techniques on structured datasets
- Built data preprocessing pipelines to clean, transform, and prepare data for model training and evaluation
- Created Python scripts for model training, testing, and performance evaluation using standard ML workflows
- Worked on model optimization and debugging, improving overall code quality and performance

- Gained hands-on experience in end-to-end ML workflows, including data handling, model building, and validation

PROJECTS On-Device RAG Application with Gemma LLM and Orama Vector Database

- Developed an on-device AI system using Python, implementing Retrieval-Augmented Generation (RAG) to retrieve relevant information and generate accurate responses
- Used Gemma LLM for local inference, enabling offline, prompt-based response generation
- Implemented semantic search using Orama vector database and embeddings, converting text into numerical vectors for efficient similarity-based retrieval
- Built a preprocessing pipeline (Python) including text cleaning, chunking, embedding generation, and indexing for better search performance
- Combined retrieval + LLM generation, improving response quality by providing relevant context to the model
- Designed the system for local deployment, ensuring data privacy, reduced latency, and real-time performance

Technologies: Python, Gemma LLM, RAG, Orama Vector DB, Embeddings, NLP

Autism Detection from 3D Brain MRI using Deep Learning

- Developed a deep learning pipeline using Python to classify 3D MRI brain scans for autism detection
- Performed MRI preprocessing (skull stripping, normalization) using medical imaging libraries to improve data quality
- Implemented CNN models using TensorFlow/Keras for feature extraction and classification of brain images
- Applied SHAP for model interpretability, identifying important brain regions influencing predictions
- Evaluated model performance using standard metrics to ensure reliability

Technologies: Python, TensorFlow, Nilearn, ANTs, DIPY, SHAP

Intelligent Resume Screening System using Machine Learning

- Developed a resume screening system using Python, automating candidate selection by matching resumes with job descriptions
- Applied NLP techniques (tokenization, TF-IDF) to convert resume text into numerical features for analysis
- Built a machine learning-based classification system to categorize resumes into relevant job roles based on content
- Implemented cosine similarity scoring, comparing resume and job description vectors to rank candidates by relevance
- Performed data preprocessing and feature extraction using Pandas and Scikit-learn to improve model performance

Technologies: Python, Scikit-learn, Pandas, NLP, TF-IDF, Cosine Similarity, Streamlit

LabourJet – Smart Labour Availability Web Application

- Developed a web-based service booking platform using Python, connecting users with available workers through a structured system
- Designed backend logic and workflows in Python, handling user requests, job assignments, and service management
- Implemented database management using MySQL, storing user data, worker details, and booking information efficiently
- Built separate modules for user, admin, and worker roles, enabling smooth interaction and system control
- Completed the full development lifecycle, from system design and database schema creation to implementation and testing

Technologies: Python, MySQL, Backend Development